

Hong-Ming Chiu

PhD Student at UIUC ECE

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RESEARCH INTERESTS

I am broadly interested in **optimization** and **machine learning**. My current research focuses on developing theoretical foundations for **trustworthy machine learning** and practical algorithms for **neural network robustness verification**.

EDUCATION

University of Illinois Urbana-Champaign (UIUC)

Doctor of Philosophy – Dept. of Electrical and Computer Engineering (ECE)

Advisor: Richard Yi Zhang.

Champaign, IL

Aug. 2021 – Present

National Yang Ming Chiao Tung University (NYCU)

Bachelor of Science in Electronics Engineering

Hsinchu, Taiwan

Sep. 2016 – Jun. 2020

RESEARCH & WORK EXPERIENCE

UIUC Coordinated Science Lab

Research Assistant (half-time). Supervisor: Prof. Richard Yi Zhang.

Champaign, IL

Aug. 2021 – Present

NYCU Artificial Intelligence and Multimedia Lab

Research Assistant (full-time). Supervisor: Prof. Wen-Huang Cheng.

Hsinchu, Taiwan

Jul. 2020 – Feb. 2021

USC Signal Transformation, Analysis and Compression Group

Summer Research. Supervisors: Prof. Antonio Ortega & Prof. Carrson C. Fung

Los Angeles, CA

Jul. 2019 – Aug. 2019

PUBLICATIONS (Papers and source code are available at <https://hong-ming.github.io/#mypublications>)

- [1] **Hong-Ming Chiu**, Haoyu Li, Huan Zhang, Richard Y. Zhang. “Affine-Image Propagation for Tight and Scalable ℓ_2 Neural Network Verification.” 2026 Advances in Neural Information Processing Systems.
- [2] **Hong-Ming Chiu**, Hao Chen, Huan Zhang, Richard Y. Zhang. “SDP-CROWN: Efficient Bound Propagation for Neural Network Verification with Tightness of Semidefinite Programming.” 2025 International Conference on Machine Learning. (Selected as Spotlight, one of 313/12107 submissions.)
- [3] **Hong-Ming Chiu** and Richard Y. Zhang. “Well-conditioned Primal-Dual Interior-point Method for Accurate Low-rank Semidefinite Programming.” Preprint.
- [4] Gavin Zhang, **Hong-Ming Chiu** and Richard Y. Zhang. “Fast and Accurate Estimation of Low-Rank Matrices from Noisy Measurements via Preconditioned Non-Convex Gradient Descent.” 2024 International Conference on Artificial Intelligence and Statistics.
- [5] **Hong-Ming Chiu** and Richard Y. Zhang. “Tight Certification of Adversarially Trained Neural Networks via Nonconvex Low-Rank Semidefinite Relaxations.” 2023 International Conference on Machine Learning.
- [6] Gavin Zhang, **Hong-Ming Chiu** and Richard Y. Zhang. “Accelerating SGD for Highly Ill-Conditioned Huge-Scale Online Matrix Completion.” 2022 Advances in Neural Information Processing Systems.
- [7] **Hong-Ming Chiu**, Carrson C. Fung and Antonio Ortega, “Graph Learning and Augmentation Based Interpolation of Signal Strength for Location-Aware Communications,” 2020 European Signal Processing Conference.
- [8] **Hong-Ming Chiu**, Kuan-Chih Lin and Tian-Sheuan Chang, “Run Time Adaptive Network Slimming for Mobile Environments,” 2019 IEEE International Symposium on Circuits and Systems.

SELECTED INVITED TALKS

On the Complexity of Second-Order Methods for Semidefinite Least-Squares Problems with Chordal Sparsity

INFORMS Annual Meeting, San Francisco, CA. November 1, 2026.

SB37 – Recent Advances in Linear and Semidefinite Programming.

SIAM Conference on Optimization, Edinburgh, UK. June 5, 2026.

MS388 – Advances in Modern Conic Optimization.

INFORMS Optimization Society Conference, Atlanta, GA. March 21, 2026.

SatB09 – Advances on Convex and Conic Optimization II.

Fast and Accurate Estimation of Low-Rank Matrices from Noisy Measurements via Preconditioned Non-Convex Gradient Descent

INFORMS Annual Meeting, Atlanta, GA. October 28, 2025.

TC34 – Advanced Optimization Methods in Data Science.

Well-Conditioned Primal-Dual Interior-Point Method for Accurate Low-Rank Semidefinite Programming

International Conference on Continuous Optimization, Los Angeles, CA. July 21, 2025.

Session 1S – Advances in Semidefinite Programming: Relaxations, Hierarchies, and Optimization Techniques.

International Symposium on Mathematical Programming, Montreal, Canada. July 24, 2024.

WA208 – Low-Rank Semidefinite Optimization, Algorithms, Software and Applications.

Neural Network Certification via Nonconvex Low-Rank Semidefinite Relaxations

SIAM Conference on Optimization, Seattle, WA. June 3, 2023.

MS331 – Algorithms and Applications of Large-Scale SDPs.

PROFESSIONAL SERVICE

Conference Reviewer

ICML 2025, 2023.

NeurIPS 2026, 2024, 2023, 2022.

ICLR 2026, 2024.

AISTATS 2024, 2023.